Array 32 (2026) 101221

Contents lists available at ScienceDirect

Array

journal homepage: www.elsevier.com/locate/array

SnakeSAM: A topology-preserving foundation model for medical curvilinear segmentation

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ARTICLE INFO

Keywords:
Foundation models
Dynamic Snake Adapter
Path-field prompting
CurvX-582K dataset

ABSTRACT

Foundation models, epitomized by the Segment Anything Model (SAM), have achieved unprecedented success in zero-shot generalization. However, their performance encounters a bottleneck in the segmentation of curvilinear structures in medical images, such as blood vessels and nerves. We identify that this limitation primarily stems from SAM's inherent prompt-based paradigm, which is optimized for blob-like objects but lacks the inductive bias necessary to capture fine, elongated geometries. To address this, we propose SnakeSAM, a framework specifically designed for curvilinear structure segmentation. Specifically, we introduce a Dynamic Snake Adapter (DSA) into the pre-trained hierarchical encoder. The DSA recalibrates frozen features using high-frequency boundaries and curvilinear structural priors, thereby enhancing the model's sensitivity to subtle geometric details. Furthermore, we design a Path-Field Prompting (PFP) strategy that transforms sparse prompts into continuity-aware representations by modeling the latent spatial dependencies between prompt points. To ensure topological consistency, we autonomously generate corrective prompts by identifying topological breaks and high-uncertainty regions in initial predictions, bridging discontinuities during multi-round inference. We validate our method on a constructed large-scale benchmark dataset (CurvX-582K) comprising 101k images and 582k masks. Experimental results demonstrate that SnakeSAM consistently outperforms existing state-of-the-art foundation models and achieves performance comparable to task-specific expert models. Cross-dataset and zero-shot evaluations further highlight its utility and generalization potential in real-world clinical applications.

1. Introduction

Medical image segmentation serves as the cornerstone of intelligent medical visual analysis, aiming to precisely extract quantitative parameters of key structures from complex anatomical backgrounds [1]. It provides critical decision support for precision diagnosis, interventional surgical planning, and the assessment of pathological evolution. In clinical practice, anatomical structures can generally be categorized into two morphological classes: Blob-like Structures (e.g., lung lobes, liver, tumors), characterized by closed boundaries and compact geometric centroids; and the more demanding Curvilinear Structures (e.g., multi-level vascular trees, nerve bundles, bronchi). In contrast to blob-like objects, curvilinear structures exhibit extreme aspect ratios, highly branching topological evolution, and drastic cross-scale morphological variations, making their precise representation in medical images exceptionally challenging [2,3]. Achieving accurate segmentation of curvilinear structures holds inestimable clinical value. In ophthalmology, for instance, the fine morphology of retinal vessels serves as a vital biomarker for diagnosing systemic diseases such as diabetic retinopathy, glaucoma, and hypertension. Microscopic morphological changes in the optic nerve and inner retinal layers provide a window for observing degenerative lesions in the central nervous system. Furthermore, the geometric characteristics of retinal microvasculature often map to the health status of distal vascular systems, such as the heart and kidneys [4,5]. Therefore, developing segmentation models capable of handling complex topological variations is of significant scientific importance for advancing precision medicine, particularly in automating diagnosis and treatment in ophthalmology and cardiovascular fields.

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https://doi.org/10.1016/j.array.2026.101221
Received 22 May 2026; Received in revised form 18 August 2026; Accepted 2 September 2026
Available online 9 September 2026
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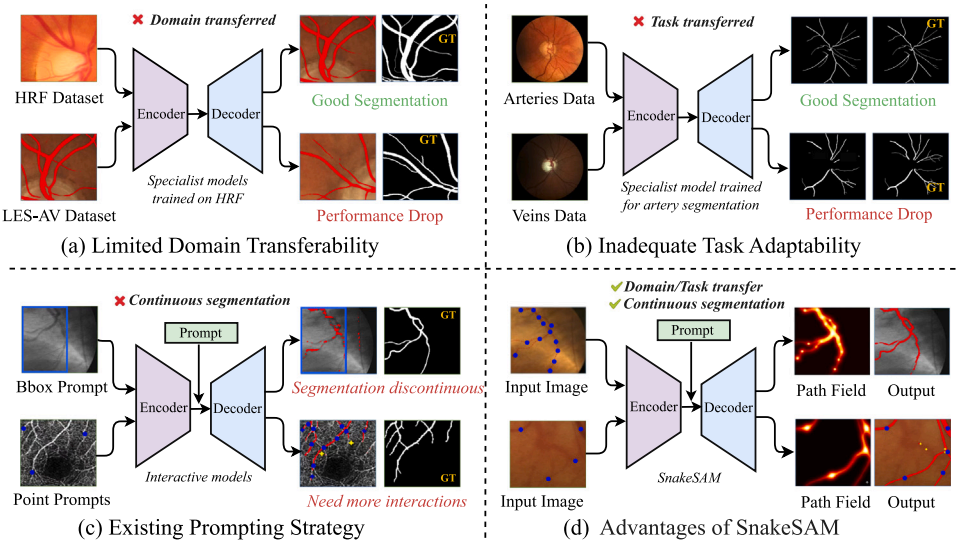


Fig. 1. Advantages of SnakeSAM in curvilinear structure segmentation tasks compared to specialist models and interactive models.

Traditional methods for curvilinear structure segmentation largely relied on hand-crafted geometric filters or heuristic expert rules [6,7]. While these methods offer a degree of interpretability, their robustness often falls short when dealing with low-contrast or noisy clinical data. With the rise of deep convolutional neural networks (CNNs) and Transformers, deep learning-based approaches have significantly improved segmentation accuracy by designing complex attention mechanisms [8–10], multi-scale fusion modules [11,12], and topology-preserving loss functions [2,13]. However, despite their superior performance on specific tasks, these specialist models face non-negligible systematic limitations: (1) Limited Domain Transferability: Specialist models often overfit to specific data distributions. As shown in Fig. 1(a), a high-performance U-Net pre-trained on the HRF dataset suffers catastrophic performance degradation when transferred to the LES-AV dataset, despite the similarity in distribution. (2) Inadequate Task Adaptability: Existing specialist models lack flexibility. As illustrated in Fig. 1(b), even models trained on large-scale fundus vessel datasets often require architectural redesign or retraining from scratch when facing derivative tasks, such as fine-grained artery/vein classification, making rapid cross-task adaptation difficult.

Recently, the emergence of foundation models, such as the Segment Anything Model (SAM), has introduced a novel interactive paradigm for image segmentation [14]. Leveraging generalized knowledge learned from over 1 billion masks, SAM achieves remarkable zero-shot segmentation performance via clicks (points), bounding boxes, or natural language prompts. However, within the specific context of medical imaging, particularly for slender curvilinear structures, SAM's performance is suboptimal. Given that medical image annotation relies heavily on expert experience and data is scarce, transferring SAM to the medical domain (e.g., MedSAM [15], SAM-Med2D [16]) has become a mainstream trend. Nevertheless, our research identifies a severe geometric mismatch in existing prompting strategies when handling curvilinear structures: for slender and highly tortuous vessels or nerves, conventional centroid clicks or rectangular box prompts fail to effectively cover subtle branches, frequently leading to severe disconnections or false positives in the segmentation results (as shown in Fig. 1(c)).

To address these challenges, we propose SnakeSAM, a curvilinear structure segmentation framework designed to endow foundation models with topological awareness. We revisit the failure mechanisms of foundation models on slender geometries and specifically embed a Dynamic Snake Adapter (DSA) into the pre-trained encoder. By leveraging high-frequency boundary enhancement and geometric prior calibration, the model gains sensitivity to extremely subtle branches. To overcome the shape coverage limitations of traditional prompting strategies, we design Path-Field Prompting (PFP), which transforms discrete interactions into continuity-aware representations, significantly deepening the model's understanding of tortuous topologies. SnakeSAM also incorporates an adaptive topological repair loop that autonomously identifies breaks in masks and generates corrective prompts, ensuring anatomical integrity during multi-round inference. As shown in Fig. 1(d), SnakeSAM overcomes the limitations of task-specific experts, achieving superior generalization performance without the need for retraining on new datasets, thus providing a robust interactive tool for precision medicine in large-scale clinical applications. The main contributions of this paper are summarized as follows:

- We propose SnakeSAM, a foundational model framework designed for interactive segmentation of curvilinear structures in medical imaging, leveraging fine-tuning with a dynamic snake adapter.
- We introduce a novel path-field prompting strategy, which effectively transforms sparse, discrete user interactions into continuous, connectivity aware representations, enhancing the model's ability to capture long range spatial dependencies in tortuous structures.
- We have constructed the CurvX-582K benchmark dataset, which includes 101k images and 582k high-quality masks, providing a powerful resource for advancing research on foundation models in complex morphological analysis.
- SnakeSAM consistently demonstrates superior zero-shot generalization capabilities across various clinical tasks, effectively bridging the performance gap between generalist foundation models and task-specific expert systems.

2. Related work

Our work draws upon and extends three primary areas of research: curvilinear structure datasets, specialized segmentation models, and the adaptation of the SAM for medical imaging.

2.1. Curvilinear structure datasets

Datasets focusing on curvilinear structures, such as blood vessels [17–21], nerves [22], and lumina are pivotal for clinical decision-making, particularly in ophthalmology and cardiology [23,24]. However, the current landscape of public medical datasets is fragmented

and limited in scale. Retinal Vessel Datasets: Classic datasets like CHASEDB1 [19], DRIVE [20], HRF [17], STARE [21], IOSTAR [25], and ORVS [26] provide pixel-level annotations for retinal vessels but are severely constrained by sample size (typically < 50 images). Furthermore, task definitions vary significantly; for instance, LES-AV [18] requires fine-grained artery/vein differentiation. OCTA-500 [24] focus on en-face projections of Optical Coherence Tomography Angiography, labeling features like the Foveal Avascular Zone (FAZ), capillaries, and larger vessels. In contrast, the CORN [22] dataset targets corneal nerves, which exhibit lower contrast and poorer continuity than retinal vessels. For X-ray coronary angiography, DCA1 [23] provides 134 images annotated by experts. Despite their value, the small scale and lack of diversity in individual datasets hinder the development of generalizable segmentation methods. Existing datasets are insufficient for training robust foundation models. To address this, we aggregate and standardize these resources to construct a large-scale, diverse benchmark, aiming to propel the advancement of universal curvilinear structure segmentation.

2.2. Curvilinear structure segmentation models

Deep learning approaches for curvilinear segmentation generally fall into two categories: architectural innovations and training strategy optimization. Early works focused on enhancing the U-Net architecture [8]. CENet [27] and Curv-Net [28] introduced multi-scale feature extraction to preserve spatial details, while R2U++ [29] integrated recurrent neural networks (RNNs) for retinal vessel segmentation. To capture long-range dependencies, attention mechanisms and Transformers have been widely adopted. CS2-Net [22] utilizes self-attention for hierarchical representation, while TransUNet [10], SwinUNet [9], and SegNetr [12] combine CNNs with Transformers to overcome the limited receptive fields of convolution operations. Recent specific adaptations include TCU-Net [30] and OCT2Former [31], which tailor Transformer blocks for vascular connectivity. Beyond architecture, specialized loss functions and supervision strategies have proven effective. Lin et al. demonstrated the efficacy of scribble-based weak supervision and synthetic data pre-training [2]. While these specialist models achieve high performance on specific benchmarks, they suffer from poor transferability. Their rigid architectures and training paradigms often fail to generalize to unseen datasets or varying clinical tasks without retraining.

2.3. SAM in medical image segmentation

As a foundation model for promptable segmentation, the Segment Anything Model (SAM) has demonstrated strong zero-shot generalization through large-scale pre-training. However, its limited exposure to medical images has motivated extensive efforts to adapt SAM to the medical domain. MedSAM [15] reduces the domain gap by fine-tuning SAM on a large-scale medical image corpus and primarily adopts bounding boxes to specify anatomical targets. SAM-Med2D [16] further extends SAM to diverse 2D medical segmentation tasks through large-scale medical training and lightweight adaptation, while retaining the conventional point- and box-prompting paradigm. Other studies have explored parameter-efficient adaptation strategies, such as Low-Rank Adaptation (LoRA) and lightweight adapters, to inject medical-domain knowledge into the image encoder [32–34]. In addition, uncertainty-aware box generation and automatically generated point prompts have been investigated to reduce the burden of manual interaction.

Despite their effectiveness in bridging the appearance and semantic gap between natural and medical images, most existing medical SAM adaptations treat anatomical targets as generic segmentation objects. Their primary focus is medical-domain adaptation, parameter-efficient fine-tuning, or prompt automation, rather than the explicit modeling of curvilinear geometry and topology. Consequently, conventional bounding boxes may introduce substantial background regions for elongated structures, whereas independently encoded point prompts provide limited information about the spatial relationships and connectivity among different branches. These limitations are particularly pronounced for slender and highly tortuous structures, such as retinal vessels, coronary arteries, and corneal nerves. SnakeSAM addresses this complementary geometric and topological problem rather than merely performing another medical-domain adaptation of SAM. Specifically, it introduces topology-oriented inductive biases at three levels. First, the DSA enhances the frozen hierarchical image encoder with deformation-aware tubular feature extraction, improving its sensitivity to thin and curved structures. Second, PFP performs relational reasoning among multiple foreground clicks and converts them into a connectivity-aware prompt representation, instead of treating each click as an independent spatial anchor. Third, the recursive refinement mechanism feeds the previous prediction back to the decoder and uses corrective prompts to progressively reduce segmentation discontinuities. Therefore, compared with MedSAM, SAM-Med2D, and other parameter-efficient SAM adaptations, SnakeSAM differs not only in its training data or adaptation module, but also in its explicit modeling of curvilinear geometry, inter-prompt relationships, and iterative topological refinement.

3. Materials and methods

As shown in Fig. 2, SnakeSAM is built upon the standard SAM architecture, aiming to endow the foundation model with topology-aware capabilities for slender curvilinear structures. The framework consists of three core components working in synergy: (1) Topology-Aware Hierarchical Encoder: We adopt SAM2's Hiera-tiny [35] as the backbone network. Distinct from original full-parameter fine-tuning, we embed Dynamic Snake Adapters (DSA) into the four stages of Transformer blocks. This recalibrates frozen pre-trained features, thereby enhancing sensitivity to curvilinear structures while preserving the generalization ability of the large model. (2) Path-Field Prompt Encoder: We design a Path-Field Prompting strategy that encodes user's sparse clicks into latent Path-Fields with directionality and connectivity, explicitly injecting continuity priors of anatomical structures into the model. (3) Topologically Recursive Mask Decoder: The decoder fuses image embeddings and prompt embeddings to generate the initial mask. During the training phase, the model simulates the user's iterative interaction process by automatically identifying topological break points and high-uncertainty regions in the previous prediction, generating correction prompts, and feeding them back to the encoder.

3.1. Topology-aware hierarchical encoder

Backbone and Efficient Adaptation. To achieve an optimal balance between computational efficiency and multi-scale representation learning, we employ Hiera, a hierarchical Vision Transformer pre-trained based on Masked Autoencoders (MAE), as the visual backbone. Given an input image $X \in \mathbb{R}^{H\times W\times 3}$, the encoder generates feature maps $\{F_i\}_{i=1}^{4}$, where $F_i \in \mathbb{R}^{\frac{H}{2^{i+1}}\times\frac{W}{2^{i+1}}\times C_i}$. Although Transformers excel at modeling long-range dependencies, they often lack the inductive biases required to precisely delineate curvilinear and topological structures. To bridge this gap, we introduce the Dynamic Snake Adapter (DSA). Similar to other parameter-efficient fine-tuning methods, we freeze Hiera's pre-trained parameters and only update the lightweight DSA. This allows the model to adapt to tubular structures without catastrophic forgetting of general visual features.

Dynamic Snake Adapter. DSA adopts a three-branch architecture to fuse static local contexts with dynamic topological features. Formally, for the input feature map F_i, DSA aggregates features from three parallel paths:

$$x_{out} = \mathcal{F}_{fuse}\left(\text{Concat}\left[\text{Conv}_{std}(F_i), \text{DSConv}_x(F_i), \text{DSConv}_y(F_i)\right]\right) + F_i \quad (1)$$

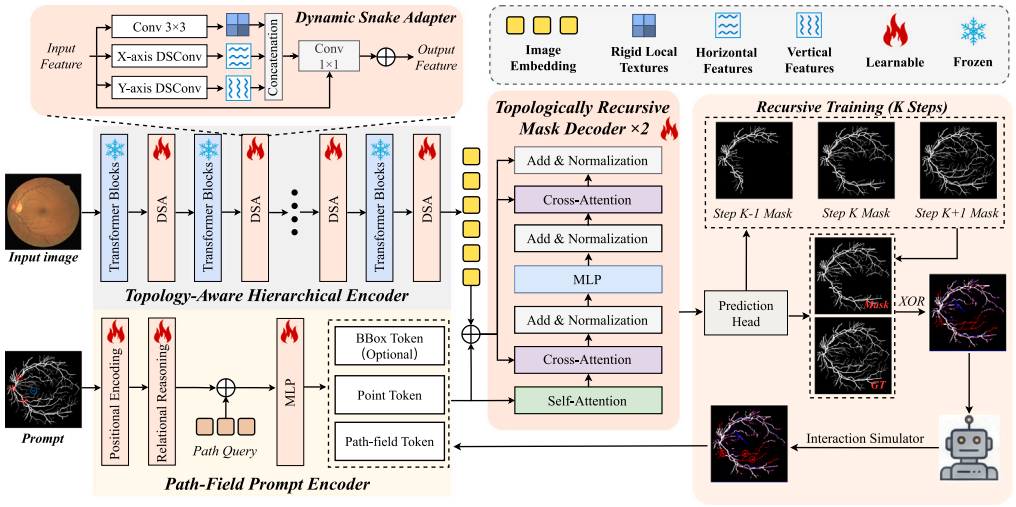


Fig. 2. Overview of the proposed SnakeSAM framework. The model consists of three core components designed to enhance topology awareness for slender structures: (1) Topology-Aware Hierarchical Encoder: We insert lightweight Dynamic Snake Adapters (DSA) into the frozen Hiera backbone. As detailed in the top-left, the DSA fuses rigid local textures (via standard convolution) with deformation-robust features (via Dynamic Snake Convolutions) to capture curvilinear geometry without forgetting pre-trained knowledge. (2) Path-Field Prompt Encoder: This module transforms sparse user clicks into a latent path-field token via relational reasoning, explicitly injecting connectivity and directionality priors. (3) Topologically Recursive Mask Decoder: Simulating human interaction, the decoder performs recursive training over K steps. It utilizes an Interaction Simulator to identify topological errors (via XOR between prediction and GT) and feeds corrective prompts back into the network to progressively refine the segmentation mask.

where Conv_{std} denotes the standard convolution used to capture rigid local textures, while DSConv_x and DSConv_y represent Dynamic Snake Convolutions [36] focused on extracting deformation-robust features along the horizontal and vertical axes, respectively. $\mathcal{F}_{fuse}$ is a 1×1 convolution for channel fusion.

We leverage DSConv to explicitly model the geometric variations of tubular structures. Unlike standard convolutions with fixed grids, DSConv introduces learnable deformation offsets. Taking the x-axis component with a kernel size of 9 as an example, the coordinate calculation for the cth step starting from the center K_i is formulated as an iterative process:

$$K_{i\pm c} = \begin{cases} (x_{i+c}, y_{i+c}) = (x_i + c, y_i + \sum_{j=i}^{i+c} \Delta y_j), \\ (x_{i-c}, y_{i-c}) = (x_i - c, y_i + \sum_{j=i}^{i-c} \Delta y_j), \end{cases} \tag{2}$$

where Σ represents the accumulation of continuous offsets Δ. The y-axis component $K_{j\pm c}$ follows a symmetrical formula. Bilinear interpolation is applied to handle fractional coordinates:

$$K(\mathbf{p}) = \sum_{\mathbf{q} \in \mathcal{N}(\mathbf{p})} \text{Bilinear}(\mathbf{q}, \mathbf{p}) \cdot x(\mathbf{q}) \tag{3}$$

Multi-Scale Feature Fusion. Although the DSA-enhanced encoder captures rich topological priors, the inherent downsampling operations of hierarchical Transformers lead to the loss of fine-grained spatial details, which are crucial for segmentation tasks. Therefore, we incorporate a Feature Pyramid Network (FPN) as the Neck [37]. First, lateral connections with 1×1 convolutions are applied to project all feature levels to a unified channel dimension d_{model} (e.g., 256) to eliminate channel discrepancies between different stages of Hiera. Subsequently, a top-down pathway is established to propagate semantic strength to high-resolution features. Specifically, high-level features are upsampled via nearest-neighbor interpolation and fused with the corresponding low-level features through element-wise addition:

$$P_i = \text{Conv}(F_i) + \text{Upsample}(P_{i+1}) \tag{4}$$

where P_i denotes the fused feature at level i. This mechanism ensures that the final feature representation encompasses both the robust semantic context from deep Transformer layers and the fine-grained topological details preserved by the DSA.

3.2. Path-field prompt encoder

Existing interactive segmentation methods primarily rely on bounding boxes (BBoxes) and click points as prompts. However, such isotropic prompts suffer from a significant geometric mismatch when dealing with tubular structures (e.g., blood vessels, nerves): BBoxes often contain excessive background noise, while discrete clicks lack the connectivity and directionality priors required to describe curvilinear topology. To address this, we propose the path-field prompt encoder. Instead of treating clicks as isolated positional information, this module models them as control nodes describing latent curvilinear trajectories, generating path-field embeddings with global connectivity by explicitly reasoning about the topological relationships between points.

Positive Embedding. Given a set of user clicks $\mathcal{P} = \{(\mathbf{c}_k, l_k)\}_{k=1}^{N}$, where $\mathbf{c}_k \in \mathbb{R}^2$ are coordinates and $l_k \in \{0, 1\}$ represents background/foreground labels. Unlike standard SAM which treats all points equally, we posit that for slender structures, only foreground points ($l_k = 1$) constitute the target's skeleton, whereas background points serve primarily to suppress false positives. Therefore, we first construct a positive sample set $\mathcal{P}^+ = \{\mathbf{c}_k | l_k = 1\}$. Subsequently, Positional Encoding (PE) is utilized to map these sparse coordinates into a high dimensional feature space:

$$\mathbf{E}_{pos} = \text{PE}(\mathcal{P}^+) \in \mathbb{R}^{M \times D} \tag{5}$$

where $M = |\mathcal{P}^+|$ denotes the number of positive samples and D is the embedding dimension.

Relational Reasoning. To recover continuous topological structures from discrete points, the model must comprehend the sequential order and dependencies among clicks (e.g., determining whether two points belong to the same vessel branch). We introduce a Transformer-based relational reasoning module [38]. Self-Attention is employed to capture long-range dependencies between point pairs:

$$\mathbf{E}_{rel} = \text{TransformerBlock}(\mathbf{E}_{pos}) \tag{6}$$

In this process, the Transformer dynamically aggregates contextual information among sparse points, such that the feature of each point is

no longer isolated but encodes its relative positional relationship within the overall topological structure.

Latent Path-Field Generation. Finally, we aggregate the relation enhanced point features into a unified path-field token. To enhance representation robustness and introduce prior knowledge, we initialize a learnable path query vector $\mathbf{q}_{path} \in \mathbb{R}^D$. We add the masked mean-pooled global point features to this query vector and perform feature remapping via a Multi-Layer Perceptron (MLP):

$$\mathbf{t}_{path} = \mathrm{MLP}\left(\mathbf{q}_{path} + \frac{1}{M}\sum_{j=1}^{M}\mathbf{E}_{rel}^{(j)}\right) \tag{7}$$

The resulting path-field embedding $\mathbf{t}_{path}$ is injected into the sparse prompt stream, concatenated with standard point/box embeddings, and fed into the decoder. This design enables the model to perceive not only location but also direction and connectivity, thereby more accurately repairing broken topological structures during multi-round interactions.

3.3. Topologically recursive mask decoder

As shown in Fig. 2, we use a lightweight mask decoder. This decoder is responsible not only for the final feature decoding but also for empowering the model with the ability to actively repair topological breaks through a recursive closed-loop training strategy that simulates the iterative interaction process of human experts.

The decoder aims to efficiently fuse image embeddings $\mathbf{F}_{img}$ with multi-modal prompt embeddings (including sparse point/box/path-field prompts $\mathbf{P}_{sparse}$ and dense mask prompts $\mathbf{P}_{dense}$). We employ a Two-Way Transformer module to establish global dependencies. Specifically, it alternately performs two key operations: (1) Prompt-to-Image Attention: using prompt embeddings as Queries and $\mathbf{F}_{fuse}$ as Keys/Values to extract specific image regions related to prompts; and (2) Image-to-Prompt Attention: conversely using $\mathbf{F}_{fuse}$ as Queries and updated prompt embeddings as Keys/Values to refine prompt features based on image context. The updated prompt tokens are fed into an MLP to dynamically predict a set of weight vectors $\mathbf{w} \in \mathbb{R}^C$. Simultaneously, the fused features $\mathbf{F}_{fuse}$ are restored to $\mathbf{F}_{up}$ via a transposed convolution upsampling layer. The final segmentation mask $\mathbf{M}$ is computed via the spatial dot product between them:

$$\mathbf{M} = \sigma(\mathbf{w}^T \cdot \mathbf{F}_{up}) \tag{8}$$

where σ is the Sigmoid activation function.

Recursive Training. Traditional single-step training often struggles to handle topological break issues common in slender structures. To this end, we employ a recursive training strategy that simulates real user interactions. The training process is modeled as a temporal sequence consisting of K iterations ($K = 5$). In the kth iteration, the model receives not only the image embeddings stored in memory but also the current set of interaction prompts $\mathcal{U}_k$ and the low-resolution prediction mask from the previous round $\mathbf{M}_{k-1}^{low}$:

$$\hat{\mathbf{M}}_k = \mathcal{D}(\mathcal{E}(\mathbf{x}), \mathrm{Embed}(\mathcal{U}_k), \mathbf{M}_{k-1}^{low}) \tag{9}$$

where $\mathcal{D}$ and $\mathcal{E}$ represent the decoder and encoder respectively.

Topological Corrections. We define the error region between the current prediction $\hat{\mathbf{M}}_k$ and the ground truth $\mathbf{M}_{gt}$ as $\mathcal{R}_{err} = \hat{\mathbf{M}}_k \oplus \mathbf{M}_{gt}$ (XOR operation). During training, the new click prompt u_{k+1} is sampled with priority from topological key points within $\mathcal{R}_{err}$ (e.g., endpoints at breaks or erroneously connected regions). Specifically, positive sample points are sampled from foreground false negative regions to connect breaks, while negative sample points are sampled from background false positive regions to separate adhesions. Through this recursive mechanism that feeds the previous prediction back into the network as a dense prior, SnakeSAM progressively learns to self-check and correct topological defects during inference.

3.4. Construction and protocol of the CurvX-582K dataset

To bridge the gap between limited clinical sample sizes and the data-hungry nature of interactive vision foundation models, we constructed CurvX-582K, a large-scale benchmark specialized for medical curvilinear structure segmentation. CurvX-582K aggregates 2181 high-resolution clinical images from eight publicly available datasets and expands them into 101,654 standardized image patches with 582,126 fine-grained instance-level masks. Below, we elaborate on the dataset construction protocol across four key dimensions as requested.

3.4.1. Data sources and composition

CurvX-582K encompasses three diverse clinical targets across four imaging modalities to ensure cross-domain topological variability. **(i) Retinal Microvasculature:** Includes CHASEDB1 [19] (28 images), DRIVE [20] (40 images), HRF [17] (45 images), LES-AV [18] (22 images), STARE [21] (20 images), and OCTA-500 [24] (300 OCTA en-face projections). **(ii) Coronary Arteries:** Sourced from DCA1 [23] (134 angiograms). **(iii) Corneal Nerves:** Sourced from CORN [22] (1592 corneal images). The complete taxonomy, target specifications, and original/processed sample counts are summarized in Fig. 3.

3.4.2. Annotation rules and provenance

All ground-truth (GT) annotations in CurvX-582K were strictly inherited from the official, expert-validated pixel-level annotations provided by the original source datasets. No manual re-delineation or artificial boundary altering was performed, thereby preserving the clinical authority of the original labels. To convert heterogeneous raw labels into a unified format suitable for interactive segmentation, we applied the following standardized annotation protocols:

(i) Binarization & Multi-Class Mapping: Multi-class annotations (e.g., artery/vein distinctions in LES-AV or FAZ/capillaries in OCTA-500) were converted into binary target masks according to the primary curvilinear structures of interest.

(ii) Instance-Level Component Disentanglement: Standard binary masks lack instance-level discrimination for interactive prompting. We algorithmically decomposed global binary masks into isolated connected components using 8-connectivity analysis. To simulate realistic multi-branch interactions without introducing non-anatomical artifacts, we synthesized multi-instance masks by randomly combining up to 3 disentangled components per crop.

3.4.3. Quality control and filtering

To prevent training instability and eliminate non-informative background noise, a rigorous two-stage quality control (QC) protocol was executed:

(i) File-Level Integrity Verification: All image–mask pairs underwent automated checksums and resolution validation to ensure zero spatial misalignment, corruption, or missing annotations.

(ii) Sparsity Aware Filtering: Curvilinear structures often occupy tiny foreground areas. However, extreme background patches contain no structural priors. We computed the foreground area ratio $R_{fg} = \frac{\text{Foreground Pixels}}{\text{Total Patch Pixels}}$ for all sliding window patches. Patches with $R_{fg} < 0.153\%$ were identified as pure background or uninformative noise and strictly filtered out.

3.4.4. Train/test split strategy

To guarantee zero data leakage between training and evaluation phases especially given the multi-scale sliding window cropping we instituted a grouped patient-level partitioning strategy:

(i) Patient-Level Grouping: Data split was conducted on the original full-sized images prior to patch extraction and instance synthesis. Images belonging to the same patient/subject were strictly constrained to the same split.

(ii) Split Ratios & Official Benchmark Preservation: For source datasets with officially established train/test partitions (e.g., DRIVE,

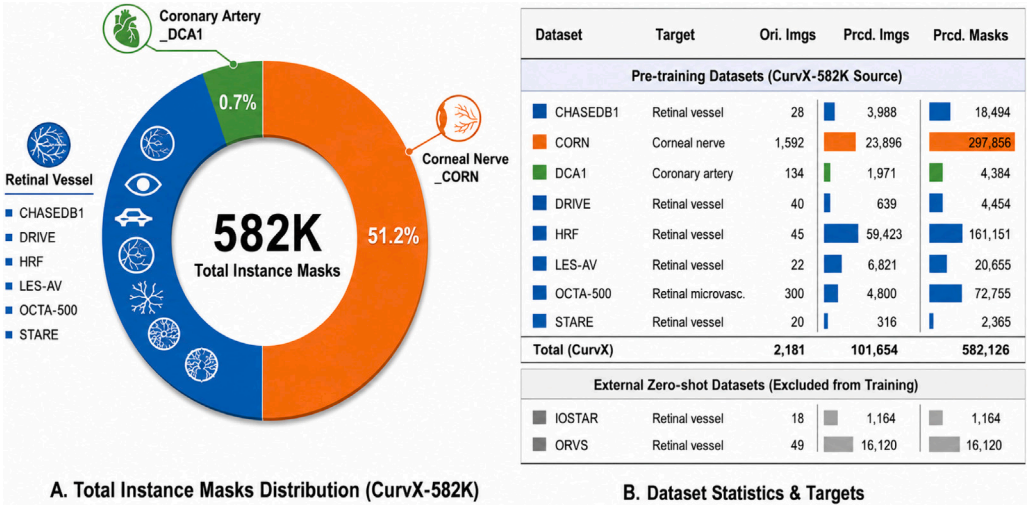


Fig. 3. Composition and statistics of the CurvX-582K benchmark. (A) The distribution of the 582 K total instance masks covers diverse anatomical targets, including Corneal Nerves, Retinal Vessels, and Coronary Arteries. (B) Detailed statistics of the data expansion process.

Table 1
Quantitative comparison between specialist and interactive models on the test set. The segmentation performance is reported as mean ± standard deviation across both overlap (Dice) and boundary distance (HD95) metrics. The best segmentation results are highlighted in bold. Computational costs are measured using an input resolution of 512 × 512 and a batch size of 1 on a single NVIDIA RTX 4090 GPU.

Method	Dice ↑		HD95 (pixels) ↓		Efficiency			
	Indep.	Joint	Indep.	Joint	Params	FLOPs	Memory	FPS ↑
Specialist Models								
UNet [8]	0.738 ± 0.007	0.653 ± 0.010	16.42 ± 0.85	23.18 ± 1.24	31.04 M	219.1 G	1.62 GB	**162.30**
TransUNet [10]	0.771 ± 0.006	0.674 ± 0.009	13.56 ± 0.72	20.45 ± 1.08	93.23 M	128.7 G	2.86 GB	32.97
FATNet [11]	0.762 ± 0.008	0.640 ± 0.011	14.80 ± 0.91	23.90 ± 1.35	28.23 M	81.6 G	2.24 GB	22.30
SwinUNet [9]	0.698 ± 0.009	0.639 ± 0.012	19.35 ± 1.12	24.73 ± 1.40	25.86 M	62.1 G	2.51 GB	26.75
EANet [39]	0.771 ± 0.006	0.675 ± 0.009	13.24 ± 0.68	19.82 ± 1.02	47.07 M	91.4 G	2.73 GB	23.39
SegNetr [12]	0.784 ± 0.005	0.708 ± 0.008	11.65 ± 0.54	16.30 ± 0.88	12.26 M	49.8 G	2.16 GB	11.54
Curv-Net [28]	0.760 ± 0.007	0.685 ± 0.009	11.20 ± 0.60	17.45 ± 0.95	**4.07 M**	**34.6 G**	**1.31 GB**	44.28
WS-DMF [40]	0.798 ± 0.005	0.718 ± 0.008	10.12 ± 0.48	15.10 ± 0.76	28.96 M	76.9 G	2.37 GB	33.52
TCSN [41]	0.794 ± 0.006	0.720 ± 0.007	9.85 ± 0.45	14.65 ± 0.70	38.64 M	101.5 G	2.71 GB	26.94
Interactive Models								
SAM-H (5 Points) [14]	0.188 ± 0.031	0.590 ± 0.023	76.40 ± 4.25	29.60 ± 2.10	635.5 M	676.9 G	7.84 GB	11.32
MedSAM (BBox) [15]	0.237 ± 0.026	0.280 ± 0.024	68.15 ± 3.80	62.30 ± 3.45	90.07 M	94.7 G	2.96 GB	40.86
SAM-Med2D (5 Points) [16]	0.226 ± 0.024	0.672 ± 0.017	71.20 ± 3.90	20.15 ± 1.25	180.1 M	139.6 G	4.31 GB	27.05
SAM-OCTA (ViT-L) [32]	0.337 ± 0.020	0.694 ± 0.014	54.30 ± 2.85	17.80 ± 1.10	308.0 M	372.4 G	5.82 GB	26.49
SnakeSAM (5 Points)	**0.879 ± 0.008**	**0.864 ± 0.006**	**4.32 ± 0.25**	**5.18 ± 0.31**	59.83 M	86.7 G	3.54 GB	29.25

OCTA-500), we strictly adhered to their official splits. For datasets without pre-defined splits, we implemented an 80%/20% split using a deterministic grouping random seed (seed = 42).

(iii) Final Statistics: The resulting merged dataset contains ∼ 81K images (460K instance masks) in the training set and ∼ 20K images (122K instance masks) in the test set. Additionally, two external benchmarks (IOSTAR [25] and ORVS [26]) were completely held out as unseen test sets to validate pure zero-shot transfer ability.

4. Results

4.1. Implementation details

We implemented the SnakeSAM model using the PyTorch framework and conducted all training on a cluster consisting of 8 NVIDIA RTX 4090 GPUs (each with 24 GB of VRAM). The model was optimized using the Adam optimizer with an initial learning rate of $1e^{-4}$, which linearly decayed to $1e^{-6}$ over 30 epochs. During data preprocessing, all input images were resized to a unified resolution of 512 × 512. To handle varying numbers of masks, we adopted a standardized sampling strategy: for each image, we randomly sampled 3 target masks. If an image contained fewer than 3 masks, we employed a random oversampling strategy to ensure consistent batch sizes. The batch size per GPU was set to 8. A key component of our training is the iterative optimization mechanism. During forward propagation, each batch undergoes 5 iterations. The initial iteration simulates user interaction by providing point prompts to generate a Path-Field. Subsequent iterations (2–5) automatically generate sample points based on previous predictions and merge them into the prompt set to update the path-field. We employed a hybrid loss function combining Focal Loss and Dice Loss with a linear weight ratio of 20:1 (i.e., $\mathcal{L} = 20\mathcal{L}_{focal} + \mathcal{L}_{dice}$). To ensure fair comparison, all baseline methods (including specialist and interactive methods) were re-trained on the same datasets using their official source codes or re-implementations. We further evaluated the computational efficiency of all methods under a unified inference protocol. Unless otherwise specified, the input resolution was fixed at 512 × 512, and the batch size was set to 1. All models were evaluated on a single NVIDIA RTX 4090 GPU using FP32 precision. The number of parameters refers to the total parameters

of the complete inference model. FLOPs were calculated for one end-to-end forward pass, including the image encoder, prompt encoder, and mask decoder where applicable. Peak GPU memory was recorded using `torch.cuda.max_memory_allocated()`. Inference speed was averaged over 200 runs after 50 warm-up iterations, with CUDA synchronization performed before and after timing. Performance was primarily evaluated using Dice and HD95, while clDice, Betti Number Error (BNE), and connectivity-related metrics were additionally employed for topology-focused analyses.

4.2. Comparison with state-of-the-art methods

We compared SnakeSAM against a series of specialist models and SAM-based interactive models across 8 datasets. Table 1 reports the aggregated quantitative results. We compared SnakeSAM against a series of specialist models and SAM-based interactive models across eight datasets. Table 1 reports the aggregated quantitative results under two complementary training protocols: independent training and joint training. In the independent setting, a separate model was trained on the training split of each dataset and evaluated on the corresponding test split. In the joint setting, a single model was trained using the pooled training splits of all eight datasets and evaluated on their corresponding test splits. These two protocols are designed to assess dataset-specific performance and robustness to heterogeneous multi-domain training, rather than out-of-distribution generalization. Unseen-domain generalization is evaluated separately in Section 4.3 through cross-dataset and external zero-shot transfer experiments.

As shown in Table 1, specialist models (e.g., UNet, FATNet) suffered significant performance degradation under the joint training protocol, with drops ranging from 2.73% to 12.51%. This decline indicates that specialist architectures struggle to reconcile distribution shifts present in heterogeneous datasets, leading to negative transfer. In contrast, SnakeSAM demonstrated superior robustness, maintaining high performance (0.864) even under joint training. This suggests that the Snake-based iterative evolution mechanism better accommodates heterogeneous source-domain distributions and alleviates negative transfer during multi-dataset training. Although standard SAM-based models (e.g., SAM-H, MedSAM) possess zero-shot capabilities, their performance on these specific tasks was suboptimal, likely due to a lack of fine-grained boundary adaptability for small or complex structures. In the independent training setting, SnakeSAM substantially outperformed the strongest SAM-based competitor, SAM-OCTA, by 54.2 percentage points (0.879 vs. 0.337), demonstrating that the integration of curvilinear geometric priors with foundation-model representations is crucial for high-precision segmentation. Consistent with the Dice results, SnakeSAM achieves the lowest HD95 under both independent and joint training (4.32/5.18 pixels), substantially outperforming TCSN (9.85/14.65 pixels) and SAM-OCTA (54.30/17.80 pixels), which further confirms its superior boundary localization accuracy.

Beyond segmentation accuracy, Table 1 provides a comprehensive comparison of model size, computational complexity, peak GPU memory, and inference throughput. SnakeSAM contains 59.83M parameters and requires 86.7G FLOPs and 3.54 GB of peak GPU memory for processing a 512 × 512 image, while maintaining an inference speed of 29.25 FPS. Compared with SAM-H, SnakeSAM reduces the parameter count by 90.6%, FLOPs by 87.2%, and peak GPU memory by 54.8%, while delivering a 2.58× higher inference speed. Compared with the ViT-L-based SAM-OCTA, SnakeSAM uses 80.6% fewer parameters, 76.7% fewer FLOPs, and 39.2% less GPU memory, while remaining slightly faster (29.25 vs. 26.49 FPS).

Although MedSAM achieves a higher inference throughput of 40.86 FPS, its segmentation performance is substantially lower than that of SnakeSAM under both the independent and joint training protocols. Similarly, several specialist models, such as Curv-Net and SegNetr, have fewer parameters or lower theoretical FLOPs, but exhibit considerably inferior segmentation accuracy and limited cross-domain adaptability. It should also be noted that FLOPs do not fully determine practical inference speed, since operator parallelism, memory access, attention implementation, and customized deformable operations can introduce different hardware utilization efficiencies. Overall, SnakeSAM is not designed to minimize computational cost alone; instead, it provides a favorable balance among segmentation accuracy, topological fidelity, computational overhead, and real-time inference capability.

Statistical significance analysis. To further determine whether the observed improvements are robust to random variation, we conducted pairwise statistical significance tests between SnakeSAM and all specialist and interactive baselines in Table 1. As shown in Fig. 4(a), SnakeSAM consistently achieves positive Dice improvements over all competing methods under both independent and joint training protocols. The absolute improvements range from 0.081 to 0.691 under independent training and from 0.144 to 0.584 under joint training. More importantly, as shown in Fig. 4(b), all pairwise comparisons remain statistically significant after Holm–Bonferroni correction ($p_{Holm} < 0.001$). These results demonstrate that the improvements achieved by SnakeSAM are consistent across repeated runs and are unlikely to be explained by random training variation.

4.3. Cross-dataset and zero-shot generalization on unseen domains

To explicitly evaluate out-of-distribution generalization, which is distinct from the multi-domain robustness assessed in Section 4.2, we established two complementary unseen-domain evaluation protocols. First, all models were trained exclusively on HRF and directly evaluated, without target-domain fine-tuning, on LES-AV, DCA1, and CORN. This setting constitutes a strict cross-dataset transfer evaluation because none of the target-domain samples were observed during training. Second, we performed external zero-shot evaluation on IOSTAR and ORVS, both of which were completely excluded from the construction and training of CurvX-582K.

As presented in Table 2, specialist models (e.g., UNet, EANet) exhibit severe brittleness under distribution shifts. While they perform reasonably well on LES-AV due to its similarity to the source domain, their performance collapses catastrophically on DCA1 and CORN. Notably, on the CORN dataset, UNet and EANet achieve Dice scores of only 0.043 and 0.018, respectively. This suggests that these models predominantly overfit to the specific textural statistics (e.g., the red–orange hue of the fundus) rather than learning the intrinsic geometric topology of curvilinear structures. When the background changes from the retina to the grayscale texture of corneal microscopy, their feature extractors fail completely. In contrast, SnakeSAM demonstrates remarkable robustness across all heterogeneous domains. On the challenging CORN dataset, SnakeSAM achieves a Dice score of 0.783, outperforming the SOTA interactive model SAM-Med2D (0.317) by a substantial margin of +46.6%. Similarly, in the cross-modality transfer to X-ray angiography (DCA1), our model maintains a high accuracy of 0.772. This validates our core hypothesis: the integration of the Dynamic Snake Adapter and Path-Field Prompting enables the model to disentangle topological priors (connectivity, tubularity) from modality-specific appearances. SnakeSAM does not merely memorize what a vessel looks like in a fundus image; instead, it comprehends the continuity of tubular structures, enabling it to generalize to unseen anatomies where traditional supervised methods and standard SAM adaptations fail.

Table 3 further shows the zero-shot test results of SnakeSAM on the IOSTAR and ORVS datasets. Compared to the current state-of-the-art interactive segmentation methods, our model is still able to achieve good segmentation performance without the need for retraining or fine-tuning. This result indicates that our SnakeSAM model has good generalization capability and can effectively handle segmentation tasks for other curved structures.

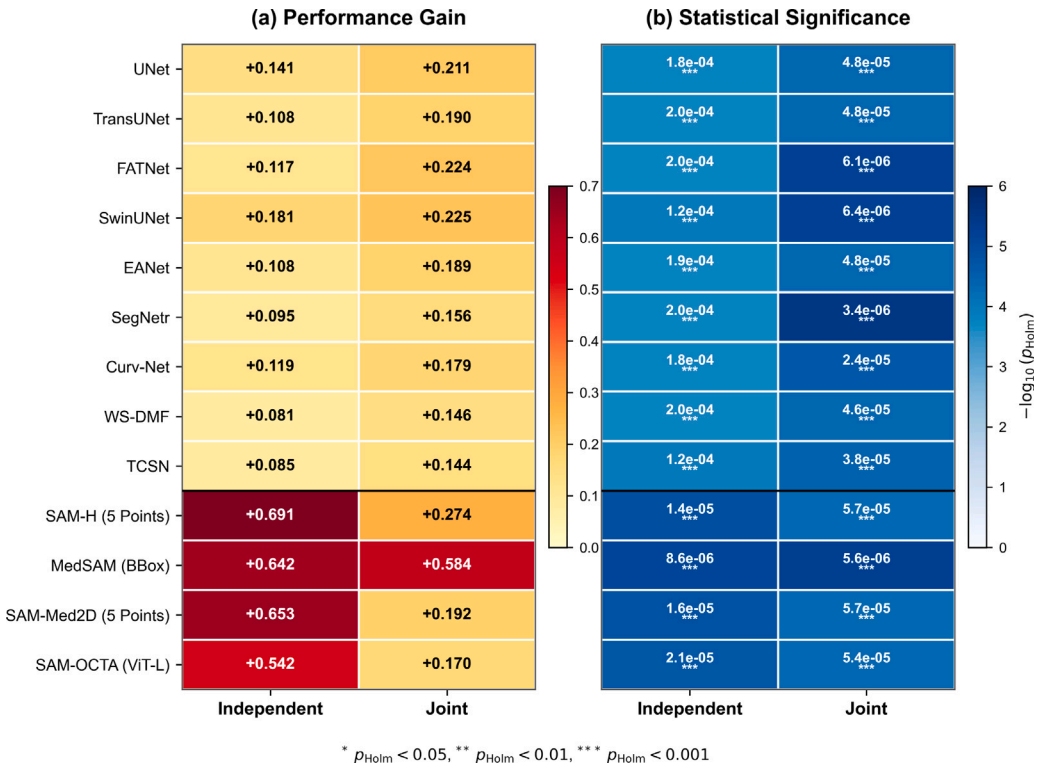


Fig. 4. Statistical significance analysis of SnakeSAM against specialist and interactive segmentation models under independent and joint training protocols.

Table 2
Cross-dataset and zero-shot generalization on unseen domains. Models were trained exclusively on the HRF dataset (Retinal) and evaluated directly on DCA1 (Cardiology), LES-AV (Retinal), and CORN (Corneal) datasets without any fine-tuning.

Method	Testing Dataset		
	DCA1	LES-AV	CORN
UNet [8]	0.107 ± 0.020	0.678 ± 0.017	0.043 ± 0.011
TransUNet [10]	0.655 ± 0.018	0.728 ± 0.016	0.133 ± 0.004
EANet [39]	0.108 ± 0.003	0.675 ± 0.010	0.018 ± 0.004
SAM-Med2D (5 Points) [16]	0.640 ± 0.005	0.697 ± 0.008	0.317 ± 0.011
SnakeSAM (5 Points)	**0.772 ± 0.006**	**0.880 ± 0.004**	**0.783 ± 0.008**

Table 3
External zero-shot evaluation on completely unseen datasets. IOSTAR and ORVS were entirely excluded from the CurvX-582K training data. SnakeSAM was directly evaluated on these two external datasets without any target-domain fine-tuning.

Method	Zero-shot Dataset	
	IOSTAR	ORVS
SAM-H (5 Points) [14]	0.542 ± 0.011	0.601 ± 0.015
SAM-Med2D (5 Points) [16]	0.657 ± 0.009	0.616 ± 0.005
MedSAM (Bbox) [15]	0.371 ± 0.010	0.422 ± 0.013
SAM-OCTA (ViT-L) [32]	0.692 ± 0.008	0.644 ± 0.007
SnakeSAM (5 Points)	**0.793 ± 0.006**	**0.735 ± 0.007**
SnakeSAM (10 Points)	**0.803 ± 0.006**	**0.776 ± 0.009**

4.4. Topological connectivity and fidelity analysis

Although the Dice Coefficient is the standard metric for volumetric segmentation, it is insensitive to topological errors in thin curvilinear structures. For instance, a single-pixel break in a long vessel may result in only a negligible decrease in DSC, yet it can severely hamper the tracing of vascular connectivity (e.g., for blood flow assessment). To rigorously evaluate the structural integrity and topological correctness of SnakeSAM, we introduce two topology-aware metrics: Centerline Dice (clDice) and Betti Number Error (BNE). Specifically, clDice quantifies the overlap between the extracted skeletons of the predicted mask and the ground truth. It penalizes segmentation results that possess high volumetric overlap but fail to capture continuous tubular centerlines. We calculate the absolute error of the first two Betti numbers as $\mathcal{E}_{betti} = |b_0 - \hat{b}_0| + |b_1 - \hat{b}_1|$, where b_0 denotes the number of connected components and b_1 denotes the number of loops (holes). A lower BNE value indicates that the model correctly preserves the topological structure.

Table 4
Topological Fidelity Comparison. We report the Centerline Dice and average Betti Number Error on the HRF test set.

Method	Dice ↑	clDice ↑	BNE ↓	Conn.Err. ↓
Specialist Models				
UNet [8]	0.709	0.715	12.4	14.2%
SegNetr [12]	0.775	0.762	8.9	9.5%
Curv-Net [28]	0.740	0.748	7.2	8.1%
Interactive Models				
SAM-Med2D [16]	0.663	0.584	15.6	21.4%
MedSAM [15]	0.310	0.215	42.1	56.8%
SnakeSAM	**0.821**	**0.892**	**2.8**	**2.3%**

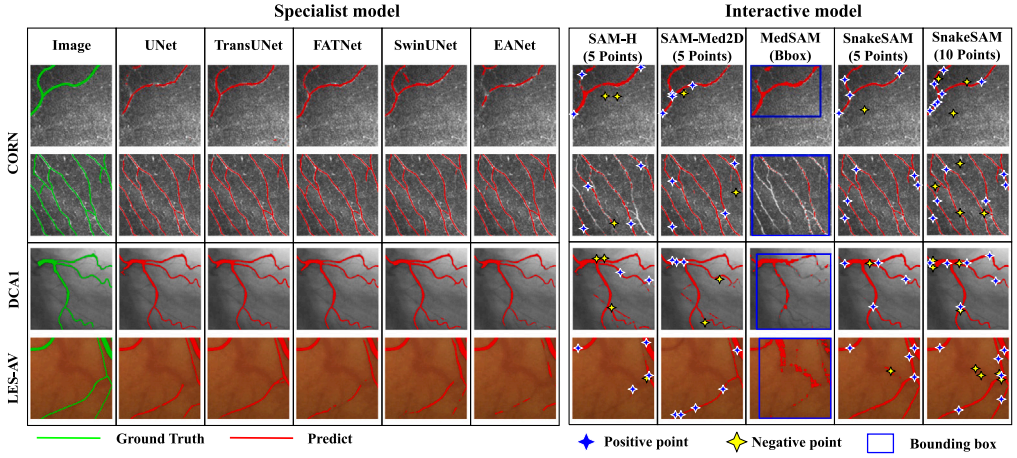


Fig. 5. Qualitative experimental results of five specialist models and four interactive models across three datasets.

Table 4 presents the quantitative results on the HRF dataset. A critical observation is the dice-topology gap in baseline methods. For instance, while SAM-Med2D achieves a reasonable Dice score (0.663), its clDice drops significantly to 0.584. This discrepancy reveals that standard SAM-based adaptations tend to segment vessels as disjointed blobs rather than continuous tubes, failing to capture the elongated geometric priors. In sharp contrast, SnakeSAM demonstrates superior topological fidelity.

SnakeSAM achieves a clDice of 0.892, surpassing the SOTA specialist model SegNetr by 13%. This indicates that our Path-Field Prompting effectively guides the model to attend to the continuous centerline of vessels rather than just local textures. Most notably, SnakeSAM reduces the BNE to the lowest BNE among the evaluated methods, at 2.8. As visualized in comparisons, baseline models often yield fragmented predictions (high b_0 error) or hallucinate non-existent loops (high b_1 error). The Recursive Mask Decoder in SnakeSAM actively detects these topological breaks and generates corrective prompts, thereby autonomously healing the disconnected components during inference.

4.5. Qualitative analysis

To intuitively evaluate the perception quality and topological correctness, Fig. 5 visualizes the segmentation results of SnakeSAM on three different datasets (LES-AV (retinal fundus), DCA1 (X-ray angiography), and CORN (corneal confocal microscopy)) compared with five representative specialized models and four interactive foundational models.

A key limitation of current SAM-based adaptive methods is clearly evident in these comparisons. As shown in Fig. 5, MedSAM's bounding box-based prompts perform poorly on curved structures. The rigid rectangular prior fails to adapt to tortuous vascular trees, resulting in coarse, blocky segmentation that either completely misses subtle branches or severely over-segments the background (e.g., in DCA1). Similarly, point-based models like SAM-H and SAM-Med2D, while improving upon MedSAM, also exhibit a speckle-like bias. They tend to segment vessels into a series of disconnected islands rather than a continuous network. This is particularly evident in the CORN dataset, where SAM-Med2D generates masks that are heavily fragmented, with numerous topological breaks, failing to capture the long-range connectivity of corneal nerves.

Specialized models (e.g., UNet, TransUNet, SwinUNet) demonstrate good performance on the LES-AV dataset, which resembles standard medical segmentation benchmarks. However, their performance significantly declines on the challenging CORN dataset. Due to the extremely low contrast and high noise levels in confocal microscopy, these models struggle to distinguish foreground nerves from background textures, leading to high false negative rates. In contrast, our model successfully delineates faint, thin nerve fibers in the CORN dataset that are nearly undetectable by other methods, confirming the effectiveness of our architecture in learning generic curve representations rather than dataset-specific textures. The most significant advantage of SnakeSAM lies in its ability to maintain anatomical continuity. Comparing the predictions on DCA1, both SnakeSAM (5 point and 10 point) accurately reconstruct complex vascular bifurcations and distal vessel termini. Unlike the comparative methods, which often introduce breaks at intersections or miss vessel endpoints, SnakeSAM preserves the coherence of the vascular tree. The visual improvement from 5 to 10 points further illustrates the effectiveness of our Path-Field prompts.

To further characterize the remaining boundary conditions, Fig. 6 presents six representative challenging cases from DCA1 and CHASEDB1. The cases were selected after full-test ranking with region- and topology-sensitive metrics followed by visual inspection, rather than solely by the lowest Dice scores. SnakeSAM preserves the main vascular trajectory in most cases; nevertheless, its performance may degrade when distal segments are barely visible or extremely thin (Fig. 6(A,D)), when dense branching, high tortuosity, or abrupt diameter changes create local topological ambiguity (Fig. 6(B,C)), and when

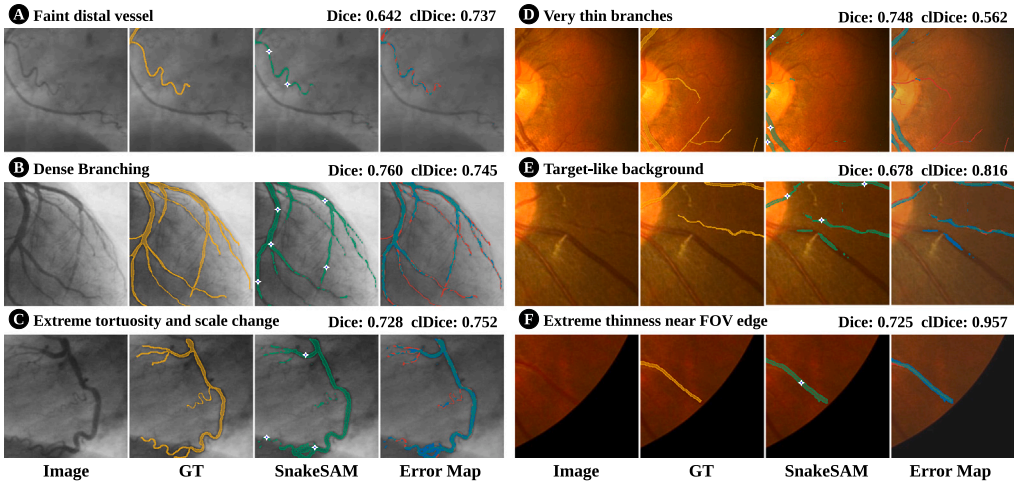


Fig. 6. Representative challenging cases of SnakeSAM on DCA1 and CHASEDB1.

target-like background structures or field-of-view boundaries interfere with delineation (Fig. 6(E,F)). Under these conditions, the model may occasionally terminate weak branches early, introduce local centerline breaks, or expand the predicted boundary toward adjacent structures. These observations indicate that, although DSA enhances curvilinear features and PFP with recursive refinement promotes structural continuity, these priors cannot fully compensate when local image evidence is absent or ambiguous; an incomplete or over-expanded intermediate mask may also be propagated during refinement. Future work will investigate uncertainty-aware endpoint recovery, scale-adaptive prompting and refinement, and explicit suppression of anatomically implausible connections.

4.6. Ablation studies

Component-wise Effectiveness Analysis. We conducted extensive ablation studies on the HRF dataset to investigate the individual contributions of each component in SnakeSAM. Unless otherwise specified, we used 5-point prompts for interaction. The SAM with Hiera-Tiny as the encoder served as the baseline model (Base), and we progressively integrated the Dynamic Snake Adapter (DSA), Path-Field Prompt (PFP), and the Recursive Training Loop (R-Loop). The reported metrics are the Dice coefficient and the centerline Dice coefficient (clDice). The experimental results are shown in Table 5.

The baseline model achieved a Dice coefficient of 0.652 and a clDice of 0.548, indicating that standard pre-trained features are inadequate for delineating fine, continuous curvilinear structures. When assessing each module individually: Integrating DSA alone (Row 2) yields a significant performance gain, with Dice increasing by 13.3% (to 0.785) and clDice rising to 0.710. This confirms that standard Transformer modules lack inductive bias for tubular geometries, whereas our DSA effectively captures high-frequency curved boundary cues; Applying PFP alone (Row 3) boosts Dice to 0.741 and clDice to 0.663 (+11.5% in clDice over Base). Unlike DSA's focus on local feature recalibration, PFP converts discrete user clicks into global connectivity priors, significantly enhancing topological continuity; Adding only R-Loop (Row 4) improves performance to 0.703 Dice and 0.612 clDice, proving that iterative feedback helps recover missed regions even without front-end feature adaptations.

When combining pairs of modules, performance further scales up consistently. Fusing DSA and PFP (Row 5) achieves 0.832 Dice and 0.825 clDice, demonstrating strong synergy between local dynamic feature alignment and global relational prompting. Similarly, combining DSA with R-Loop (Row 6) achieves 0.816 Dice and 0.791 clDice.

Table 5
Analysis of the effectiveness of different components of SnakeSAM.

Base	DSA	PFP	R-Loop	Performance	
				Dice ↑	clDice ↑
✓				0.652	0.548
✓	✓			0.785	0.710
✓		✓		0.741	0.663
✓			✓	0.703	0.612
✓	✓	✓		0.832	0.825
✓	✓		✓	0.816	0.791
✓	✓	✓	✓	**0.864**	**0.892**

Table 6
Impact of different adaptation mechanisms.

Adapter type	Dice ↑	BNE ↓
Standard Adapter [42]	0.795	8.5
LoRA (Rank=16) [43]	0.802	7.8
DCN-Adapter [44]	0.841	4.2
DSA (Ours)	**0.864**	**2.8**

Finally, integrating all three modules (Row 7) yields the top performance of 0.864 Dice and 0.892 clDice. This complete setup confirms that local geometric recalibration (DSA), continuous spatial reasoning (PFP), and active self-correction (R-Loop) work complementarily to preserve complex topological structures.

Adapter Design. To investigate the necessity of topology-specific adaptation mechanisms, we benchmark our DSA against mainstream parameter-efficient fine-tuning strategies, including Standard Adapter, LoRA, and a Deformable Convolution (DCN) variant. As shown in Table 6, while general-purpose methods like LoRA achieve respectable pixel-level accuracy (0.802 Dice), they struggle to reduce topological errors (7.8 Betti Error), primarily due to their reliance on fixed geometric grids that fail to accommodate extreme morphological variations. Although integrating DCN improves geometric flexibility, yielding a Dice of 0.841, our DSA demonstrates superior performance (0.864 Dice) and, crucially, the lowest topological error (2.8). This validates that the DSA's explicit modeling of tubular evolution via snake-like dynamic offsets is essential for capturing the continuous, meandering nature of curvilinear structures, offering a distinct advantage over generic deformation modeling.

Efficacy of Path-Field Prompting. To validate the effectiveness of our proposed Path-Field Prompting (PFP), we benchmark it against the standard SAM point embedding and an ablation variant without relational reasoning. As presented in Fig. 7(a), the standard isotropic

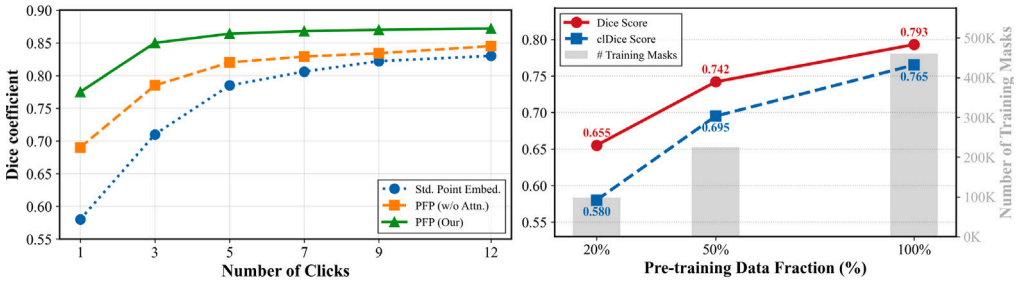


Fig. 7. Ablation studies on prompting efficiency and data scalability. (a) Comparison of segmentation performance versus the number of user clicks. (b) Impact of pre-training data scale on zero-shot generalization to the unseen IOSTAR target domain.

embedding treats user clicks as independent anchors, failing to capture the intrinsic connectivity of curvilinear structures, which results in suboptimal performance under sparse interaction (0.58 Dice at 1 click). While the ablation variant (PFP w/o Attn.) improves global context aggregation, it lacks the explicit modeling of inter-click dependencies, lagging behind the full model by 2.7% ∼ 8.5%. In contrast, our full PFP leverages self-attention to dynamically transform discrete clicks into a coherent Path-Field with directional and topological priors. Notably, with only a single click, PFP achieves a Dice score of 0.775, significantly outperforming the baseline and demonstrating its superior capability in inferring complex global topology from sparse user inputs.

Impact of Pre-training Data Scale. A key factor influencing the zero-shot generalization capability of foundational models is the scale and diversity of the pre-training corpus. To verify the effectiveness of our CurvX-582K benchmark dataset in encoding robust topological priors, we conducted a data scale study. We trained different variants of SnakeSAM using 20%, 50%, and 100% of the total pre-training data, each randomly sampled. Subsequently, we evaluated their zero-shot performance on the unseen IOSTAR target domain without any fine-tuning.

As shown in Fig. 7(b), we observed a clear positive correlation between the amount of pre-training data and downstream generalization performance. The model trained on only 20% of the data achieved a baseline Dice coefficient of 0.655. Increasing the data to 50% resulted in a significant performance improvement, with the Dice coefficient rising to 0.742. Crucially, utilizing the full 100% dataset yielded the highest performance, reaching a Dice coefficient of 0.793 and a topological metric (clDice) of 0.765. The significant performance improvements, particularly in the topological metric, highlight the importance of large-scale, diverse training data for the model to effectively grasp the complex geometric variations inherent in curved structures across different medical modalities, validating the significant value of the benchmark we have constructed.

5. Discussion

The accurate segmentation of curvilinear structures, such as blood vessels and nerve fibers, is a persistent bottleneck in medical image analysis. While foundation models like SAM [14] have established a new interactive segmentation paradigm, our extensive evaluations reveal their inherent geometric mismatch when applied to tortuous and slender anatomies. In this study, we proposed SnakeSAM, a topology-preserving foundation model that overcomes these limitations. By integrating a Dynamic Snake Adapter (DSA), Path-Field Prompting (PFP), and a recursive topological correction loop, SnakeSAM effectively bridges the gap between general visual features and the inductive biases required for curvilinear geometries.

A fundamental finding of our research is the transition from texture-biased memorization to geometry-aware reasoning. As demonstrated in our joint training and zero-shot transfer experiments (Tables 1 and 2), traditional specialist models (e.g., UNet, SegNetr) suffer from severe performance degradation when exposed to distribution shifts. For instance, models trained exclusively on retinal fundus images (HRF) failed catastrophically when applied to corneal confocal microscopy (CORN), primarily because standard convolutional or standard attention mechanisms tend to overfit to domain-specific intensity and color statistics. In contrast, SnakeSAM exhibited remarkable robustness across distinct modalities and anatomies. We attribute this generalization to the DSA, which explicitly models the meandering evolution of tubular structures, compelling the network to learn invariant geometric priors, such as continuous boundaries and tubularity, rather than relying on superficial textural cues.

Furthermore, our analysis of topological fidelity highlights a critical flaw in existing interactive models. Standard prompt embeddings treat user clicks as isolated spatial anchors. In the context of complex vascular trees, this isotropic assumption frequently leads to fragmented masks and topological hallucinations, as evidenced by the high Betti Number Errors (BNE) observed in SAM-Med2D and SAM-OCTA. Our proposed PFP module resolves this by performing relational reasoning across sparse prompts, essentially generating a latent "skeleton" that guides the model's attention along the structural continuum. Coupled with the Topologically Recursive Mask Decoder, which autonomously identifies and patches spatial breaks during inference, SnakeSAM mimics the iterative tracing behavior of human experts. This mechanism not only maximizes the centerline overlap (clDice) but also drastically reduces the manual annotation burden, requiring fewer interactions to achieve high connectivity fidelity in our experiments.

The clinical implications of SnakeSAM are substantial. In ophthalmology and cardiology, the precise extraction of contiguous vascular and neural networks is a prerequisite for quantitative biomarker analysis, such as vessel tortuosity, branching angles, and capillary dropout ratios. By providing a highly generalizable and topologically accurate segmentation tool, SnakeSAM enables large-scale, automated screening across diverse imaging modalities without the prohibitive cost of dataset-specific retraining. Additionally, the introduction of the CurvX-582K benchmark dataset, with its vast instance-level topological diversity, addresses the critical data scarcity that has historically hindered the development of foundation models for complex morphological tasks.

Despite its state-of-the-art performance, we acknowledge several limitations that warrant future investigation. First, SnakeSAM is currently designed for 2D imaging modalities. Many critical curvilinear structures, such as coronary arteries in CTA or cerebral vasculature in MRA, are intrinsically 3D. Extending the DSA and PFP mechanisms to accommodate volumetric data and 3D spatial continuity represents a crucial next step. Second, while SnakeSAM operates efficiently at 29.25 FPS, its parameter count (59.83M), though vastly smaller than standard SAM, may still pose challenges for deployment on resource-constrained edge devices (e.g., portable ultrasound or low-power fundus cameras). Future work will explore knowledge distillation and network pruning techniques to derive an ultra-lightweight variant of SnakeSAM without compromising its topological awareness.

6. Conclusion

In this paper, we presented SnakeSAM, a framework designed to reconcile the semantic generalization of foundation models with the geometric intricacies of curvilinear structures in medical imaging. We identified the inherent geometric mismatch in standard prompt-based paradigms and addressed it through two core innovations: the Dynamic Snake Adapter (DSA), which sensitizes the encoder to high-frequency structural details, and the Path-Field Prompting strategy, which transforms sparse user interactions into continuity-aware topological representations. Extensive evaluations on our newly curated large-scale benchmark (CurvX-582K) demonstrate that SnakeSAM consistently outperforms both state-of-the-art specialist models and existing interactive foundation models. Crucially, our method exhibits unprecedented zero-shot generalization capabilities across unseen modalities (e.g., transferring from retinal fundus to corneal nerves), effectively bridging the gap between generic visual understanding and precision medical analysis.

CRediT authorship contribution statement

Jingwen Li: Writing – review & editing, Writing – original draft. **Xincheng Zhang:** Supervision, Methodology. **Junlong Cheng:** Data curation. **Yujie Lu:** Methodology, Investigation. **Yanzhou Su:** Visualization. **Wei Wang:** Writing – original draft, Data curation. **Jiong Yu:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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